

INTRODUCTION TO COMPUTER SYSTEM, NETWORKS AND SECURITY

Assignment 3 – Networks Part 2 - WLAN Network and Security

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Purpose for the whole assignment 2:

- ❖ In this assignment, students will record data from a real-world wireless network and demonstrate that they can analyse it, identify its properties and potential issues. Also, students will analyse Internet traffic and identify servers, clients and protocols used. The assignment is related to Unit Learning Outcomes 5 and 6.

Purpose for the assignment 3 part 2:

TASK	BRIEF DESCRIPTION
Perform a real-world WLAN site survey.	Produce a map of part of a building that gives an overview of the wireless networks that are available, as well as an analysis of the network.

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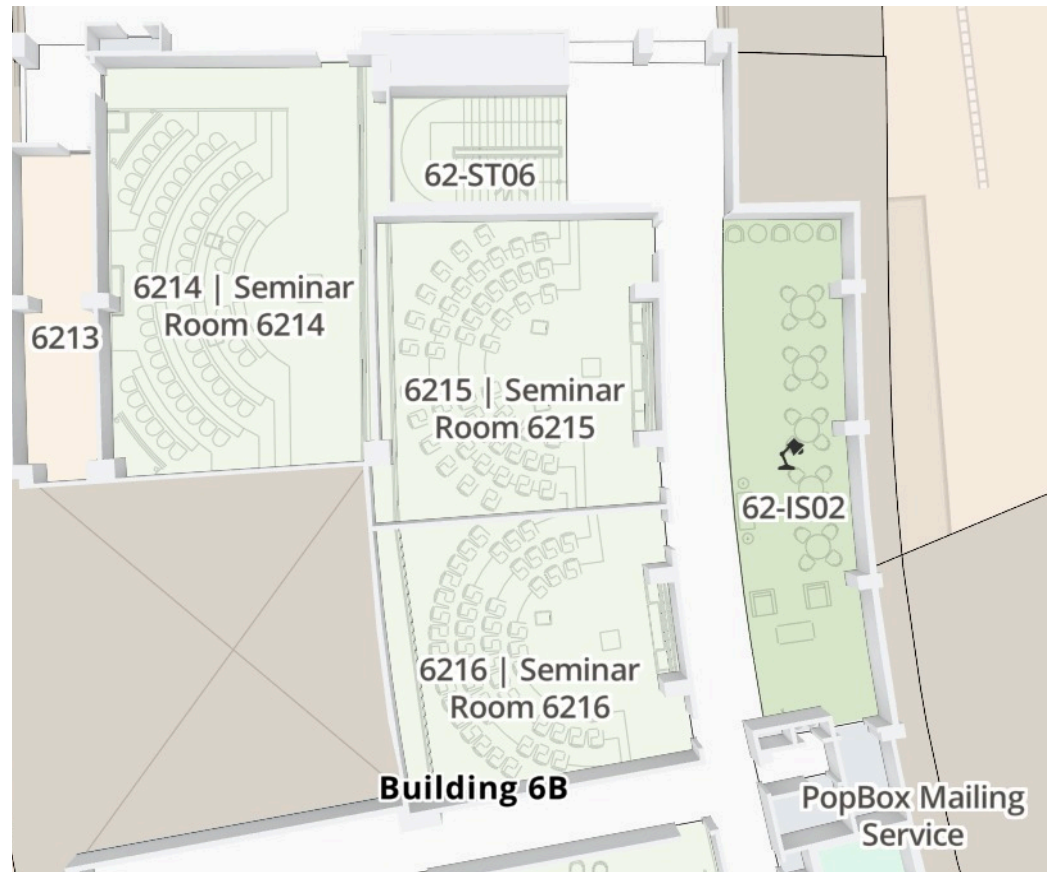
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1. Survey

1.1. Rooms data

1.1.1. Rooms chosen

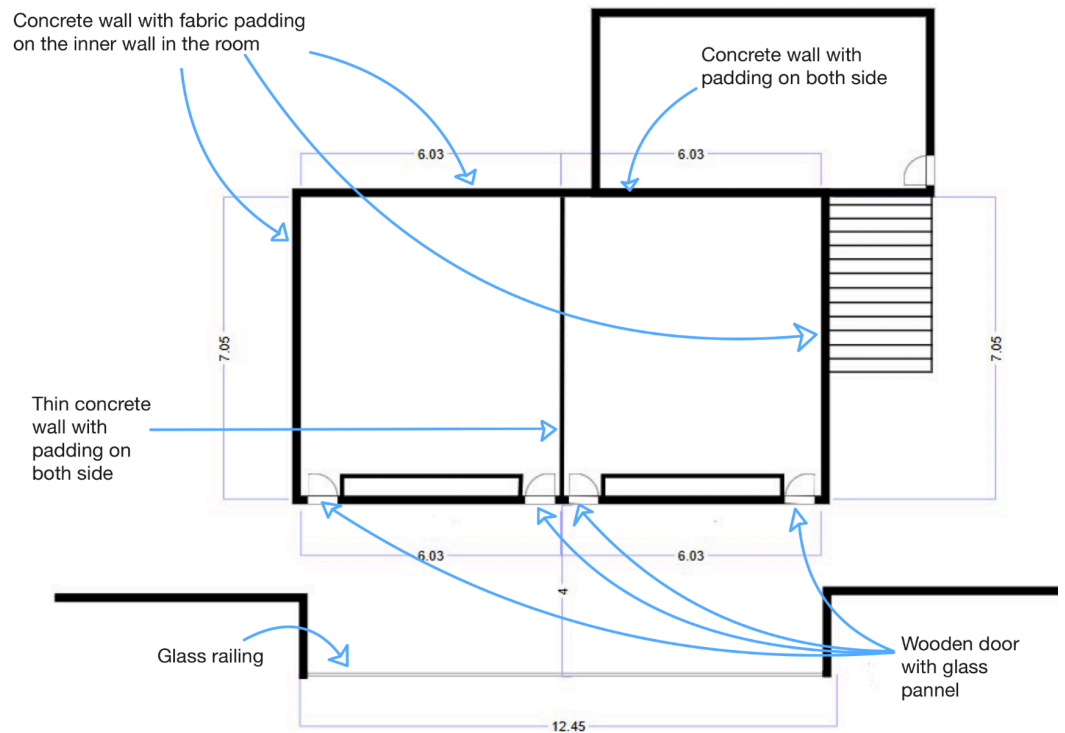
Seminar room 6215, Seminar room 6216 and the open area in front of the two room(62-IS02)



The reason I decided to survey this area is because it meets all the criteria(Having at least 3 APs) and also has a mix of closed off area and open area.

1.1.2. Floor plan

The following is a skeleton floor map of the area surveyed containing dimensions(estimated) and material types



Skeleton map of the area surveyed(Rotated 90 degrees clockwise from previous map for ease of viewing)

The total area of area is around **130 m²**

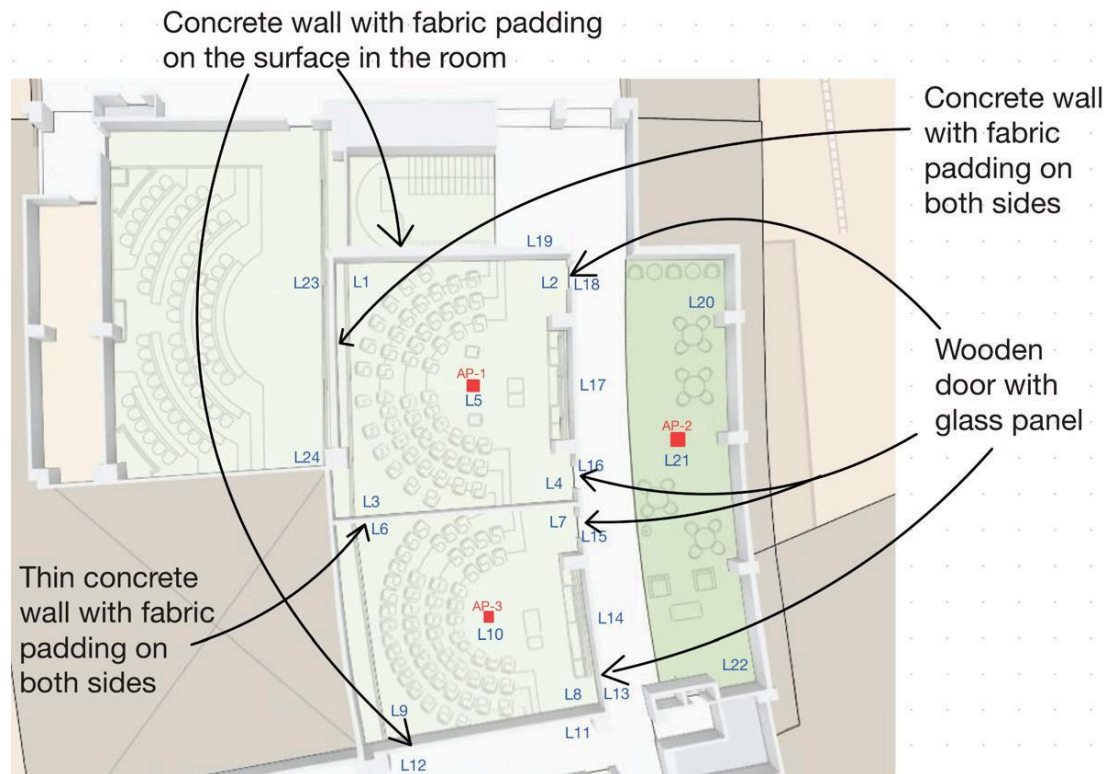
$$\text{Area} = [2(6) \cdot 7] + [12.45 \cdot 4] = 84 + 49.8 = 133.8$$

Disclaimer: Please note that the length readings are only an estimation.

1.2. Survey location & Access points

1.2.1. “Data” map

The map contains the location of the APs and where data was collected:



The access points are referenced by the notations AP-1, AP-2 and AP-3.

The survey locations are referenced by the notations L1 to L24.

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1.3. Data compilation

Data reading location	suspected AP	BSSID	name	channel	802.11 version support	band	Signal strength (dB)	security
L1	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-65	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-60	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-57	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	No data	WPA2 - Enterprise
L2	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-41	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-73	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-73	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	No data	WPA2 - Enterprise
L3	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-41	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-70	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-78	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	-74	WPA2 - Enterprise
L4	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-42	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-65	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-60	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	No data	WPA2 - Enterprise
L5	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-37	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-73	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-75	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	No data	WPA2 - Enterprise

Data reading location	suspected AP	BSSID	name	channel	802.11 version support	band	Signal strength (dB)	security
L6	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-58	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-66	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-64	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	-47	WPA2 - Enterprise
L7	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-57	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-66	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-61	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	-45	WPA2 - Enterprise
L8	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-66	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-72	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-66	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	-37	WPA2 - Enterprise
L9	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-67	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-72	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-71	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	-46	WPA2 - Enterprise
L10	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-66	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-71	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-76	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	-38	WPA2 - Enterprise
L11	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-75	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-73	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-72	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	-59	WPA2 - Enterprise

Data reading location	suspected AP	BSSID	name	channel	802.11 version support	band	Signal strength (dB)	security
L12	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-77	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-73	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-77	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	-57	WPA2 - Enterprise
L13	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-75	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-60	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-57	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	-57	WPA2 - Enterprise
L14	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-75	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-56	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-61	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	-57	WPA2 - Enterprise
L15	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-75	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-56	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-53	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	-59	WPA2 - Enterprise
L16	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-62	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-53	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-55	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	No data	WPA2 - Enterprise
L17	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-57	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-54	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-68	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	No data	WPA2 - Enterprise

Data reading location	suspected AP	BSSID	name	channel	802.11 version support	band	Signal strength (dB)	security
L18	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-56	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-54	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-68	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	No data	WPA2 - Enterprise
L19	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-62	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-79	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-66	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	No data	WPA2 - Enterprise
L20	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-64	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-58	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-65	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	No data	WPA2 - Enterprise
L21	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-66	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-43	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-41	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	No data	WPA2 - Enterprise
L22	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-76	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-55	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-65	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	No data	WPA2 - Enterprise
L23	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-80	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-76	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-75	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	No data	WPA2 - Enterprise

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Data reading location	suspected AP	BSSID	name	channel	802.11 version support	band	Signal strength (dB)	security
L24	AP-1	6C:8B:D3:4F:8A:4D	eduroam	56	n/ac	5	-56	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:AD	eduroam	36	n/ac	5	-76	WPA2 - Enterprise
	AP-2	6C:8B:D3:43:DF:A2	eduroam	11	g/n	2.4	-78	WPA2 - Enterprise
	AP-3	6C:8B:D3:5A:E6:ED	eduroam	44	n/ac	5	No data	WPA2 - Enterprise

2. Report

2.1. Channel allocation and overlap

2.1.1. From the dataset, three APs were observed

- AP-1 operating on Channel 56 (5 GHz).
- AP-2 operating on Channel 36 (5 GHz) and Channel 11 (2.4 GHz).
- AP-3 operating on Channel 44 (5 GHz).

The 5 GHz channels are well separated, with AP-1 on channel 56, AP-2 on 36, and AP-3 on 44. Since 5 GHz has non-overlapping 20 MHz channels, co-channel interference is minimal. However, AP-2's dual-band operation introduces the 2.4 GHz band on channel 11, which can potentially overlap with other nearby networks in the building (since 2.4 GHz only has three non-overlapping channels: 1, 6, 11).

Evidence of some co-channel interference is seen where AP-2's 2.4 GHz coverage overlaps with the strong 5 GHz signals from AP-1 and AP-3, though this is expected in mixed-band deployments.

2.1.2. Possible improvements

Encourage more client devices to use the 5 GHz band, where channels are less congested.

Reduce reliance on 2.4 GHz except for legacy compatibility, as most devices now support 5 GHz.

For large-scale deployments, consider reassigning 5 GHz channels with DFS (Dynamic Frequency Selection) to further minimise overlap.

2.2. Signal attenuation and obstacles

2.2.1. Physical barriers had a noticeable impact

2.2.1.1. Comparing Values(concrete walls; both normal and thin)

- L3 and L6 for AP-1 readings
Observation: At L3, signal strength is -41 dB and at L6, signal strength is -58 dB
Deduction: Thus there is a 17 dB difference between the two readings which is due to the thin concrete wall with fabric padding.
- L2 and L19 for AP-1 readings
Observation: At L2, signal strength is -41 dB and at L19, signal strength is -62 dB
Deduction: Thus there is a 21 dB difference between the two readings which is due to the concrete wall with fabric padding.
- L8 and L11 for AP-3 readings
Observation: At L8, signal strength is -37 dB and at L11, signal strength is -59 dB
Deduction: Thus there is a 22 dB difference between the two readings which is due to the concrete wall with fabric padding.

2.2.1.2. Comparing Values(Door made of wood have a glass panel)

- L2 and L18 for AP-1 readings
Observation: At L2, signal strength is -41 dB and at L18, signal strength is -56 dB.
Deduction: Thus there is a 17 dB difference between the two readings which is due to the door with glass panel.
- L8 and L13 for AP-3 readings
Observation: At L8, signal strength is -37 dB and at L13, signal strength is -57 dB
Deduction: Thus there is a 20 dB difference between the two readings which is due to the door with glass panel.

- L15 and L7 for AP2 readings
Observation: At L7, signal strength is -66, and at L15, signal strength is -56 dB.
Deduction: Thus there is a 10 dB difference between the two readings which is due to the door with glass panel.

2.2.1.3. Comparing Values For different bands(2.4 GHz and 5 Ghz)

Comparison of values will only be for AP-2 since it is the only AP transmitting both bands.

- L10 and L14
Observation:
 - ❖ At L10, signal strength for the 5GHz band is -71 dB and at L14, signal strength for the 5GHz band is -56 dB.
Attenuation = 15 dB
 - ❖ At L10, signal strength for the 2.4GHz band is -76 dB and at L14, signal strength for the 2.4GHz band is -61 dB. Attenuation = 15 dB

Deduction: Thus there is a 0 dB attenuation difference between the two bands which indicates that the signal strength across bands is not affected by the concrete wall in this scenario.

- L7 and L15
Observation:
 - ❖ At L7, signal strength for the 5GHz band is -66 dB and at L15, signal strength for the 5GHz band is -56 dB.
Attenuation = 10 dB
 - ❖ At L7, signal strength for the 2.4GHz band is -61 dB and at L15, signal strength for the 2.4GHz band is -53 dB. Attenuation = 8 dB

Deduction: Thus there is a 2 dB attenuation difference between the two bands which indicates that the signal strength across bands is slightly affected by the door with glass panel.

- L5 and L24

Observation:

- ❖ At L5, signal strength for the 5GHz band is -73 dB and at L24, signal strength for the 5GHz band is -76 dB. Attenuation = 3 dB
- ❖ At L5, signal strength for the 2.4GHz band is -75 dB and at L24, signal strength for the 2.4GHz band is -78 dB. Attenuation = 3 dB

Deduction: Thus there is a 0dB attenuation difference between the two bands which indicates that the signal strength across bands is not affected by the concrete wall in this scenario.

2.2.2. Conclusion

On all bands there are attenuation:

For concrete walls, it is about 15 to 20 dB for the thin walls and 20 to 25 dB for the normal walls.

For the wooden door with glass panel, it is around 10 to 20 dB.

As for attenuation for between the bands from AP-2, from theoretical knowledge, the 5 GHz signal should have attenuation more but the actual reading begs to differ. This may be due to the distance between reading is too small.

2.3. Coverage and dead zones

2.3.1. Survey data highlights

Strong coverage: Near L3–L5, with AP-1 as strong as -37 dBm.

Weak coverage: At L22–L23, where AP-1 dropped below -75 dBm, AP-2 at -76 dBm, and AP-3 had no data. These areas show marginal or no reliable connectivity.

Dead zone: There is no dead zone.

2.3.2. Recommendations

- Relocate or add an AP closer to the far end of the corridor (around L19–L23).
- Reorient antennas to cover shadowed areas inside seminar rooms.

2.4. Roaming and handoff potential

There is adequate overlap between AP-1, AP-2, and AP-3 in most of the surveyed areas. Signal strengths from multiple APs were typically within 10–15 dB of each other (e.g., at L6–L9). This overlap is sufficient to support seamless roaming for most devices.

2.4.1. Impact on user experience:

- For data browsing and downloads, the overlap is sufficient.
- For latency-sensitive applications (VoIP calls, video streaming), roaming may cause short disruptions in weaker overlap zones (e.g., in corridors going from L19 to L23), where signals from all APs are weak. Optimising handoff thresholds (e.g., enabling 802.11r fast roaming) would improve the experience.

2.5. Network load and bandwidth usage

Although direct bandwidth utilisation was not measured, the dual-band AP-2 is likely handling more load since it covers both 2.4 GHz and 5 GHz clients. AP-1 and AP-3 appear to provide good signal coverage, but AP-2's role as a fallback for older 2.4 GHz devices increases its utilisation.

2.5.1. Strategies to optimise performance:

- Enable band steering to push capable devices to 5 GHz.
- Implement load balancing between APs by adjusting transmit power to avoid too many clients clustering on one AP.
- Monitor usage statistics to verify whether additional APs or channel width adjustments (e.g., 40 MHz in 5 GHz) are required.

2.6. Other analysis - Interference from Non-WiFi Devices

An additional observation concerns non-WiFi interference:

The 2.4 GHz band is particularly vulnerable to interference from Bluetooth. During the survey, signal fluctuations of up to 5–7 dB were noticed in AP-2's 2.4 GHz channel at different times, suggesting possible intermittent interference.

5 GHz signals were comparatively stable, reinforcing the recommendation to prioritise 5 GHz usage.

Methodology: By comparing repeated measurements at the same location (e.g., L6 and L7) and noting inconsistent fluctuations only in the 2.4 GHz readings, interference was inferred.

Reasoning: Since walls and positioning did not change between measurements, the variation is most likely from external devices operating in 2.4 GHz.

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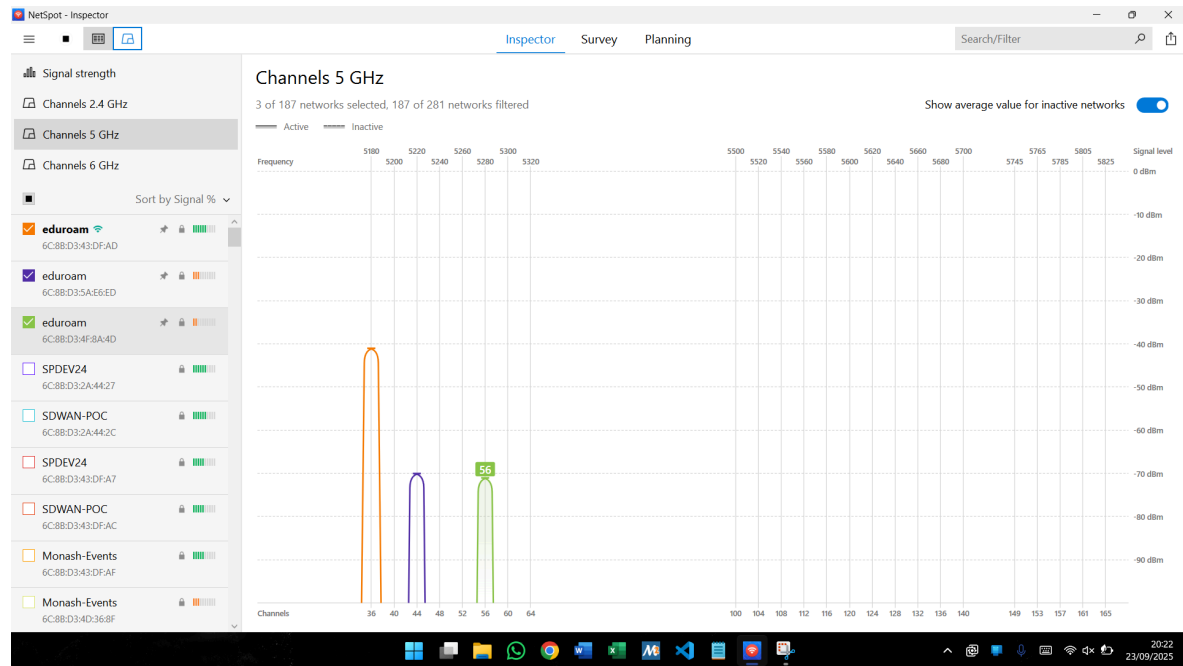
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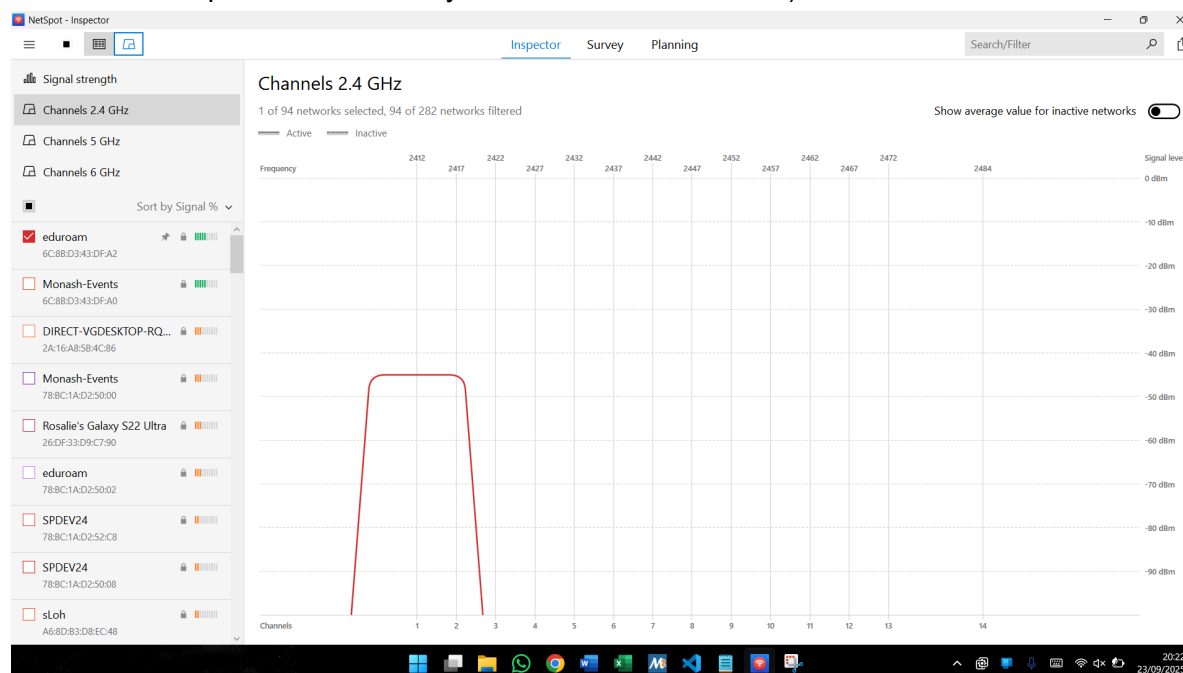
3. Appendix

3.1. Images for channel allocation and overlap

For the 5Ghz band:



For the 2.4Ghz band(Note the picture was taken at a later date from the readings, so it is showing channel 1 instead of channel 11; this does not influence the report as it is the only one on the 2.4GHz band)



—Assignment 3—

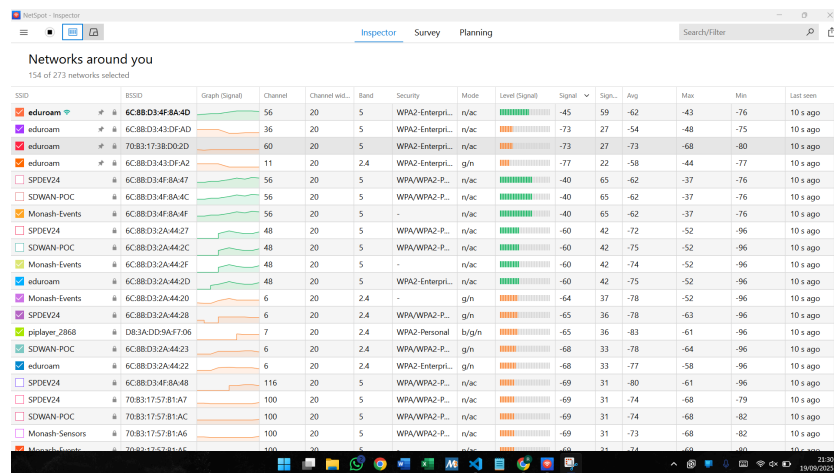
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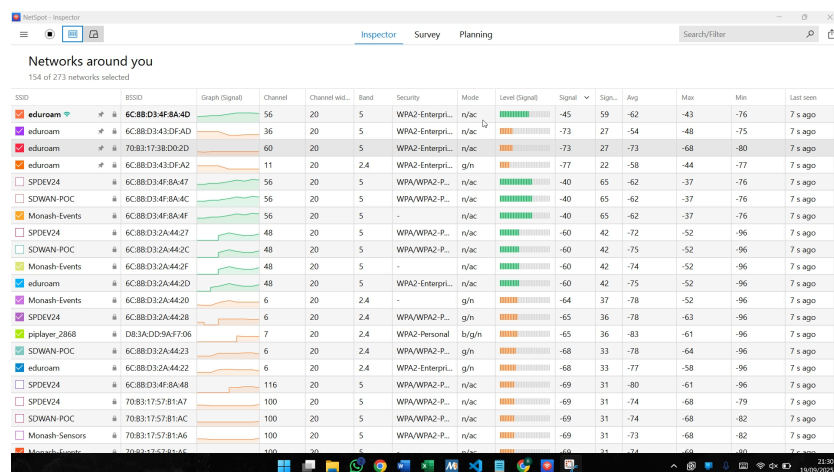
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3.2. Readings location data

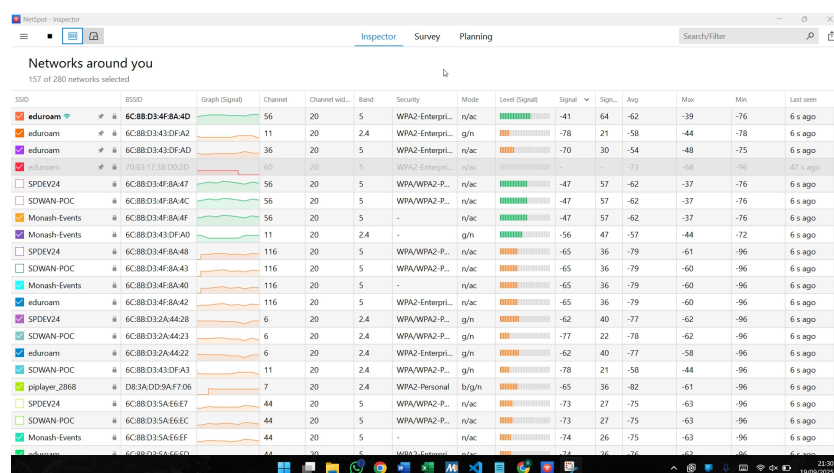
3.2.1. L1



3.2.2. L2



3.2.3. L3



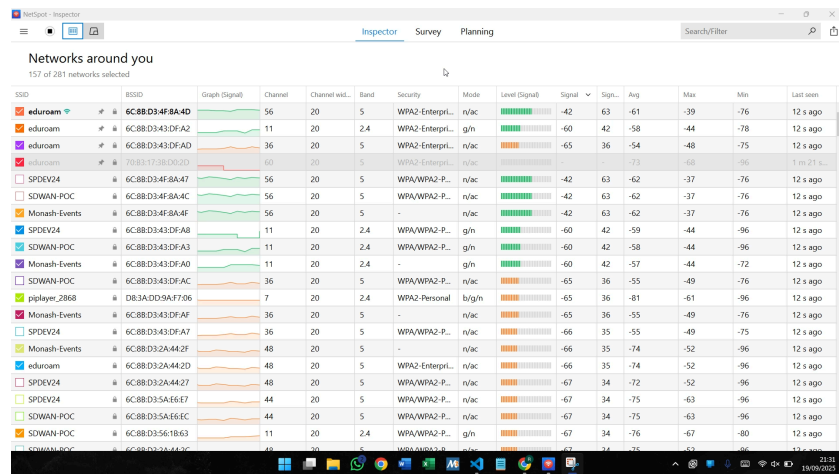
—Assignment 3—

Networks

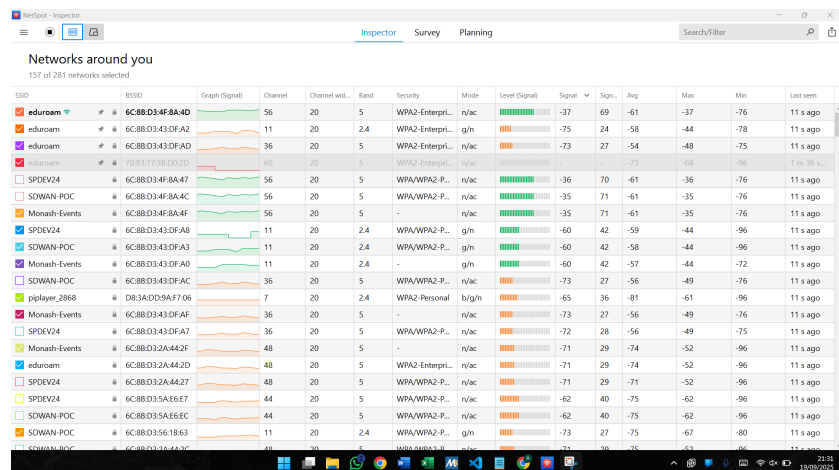
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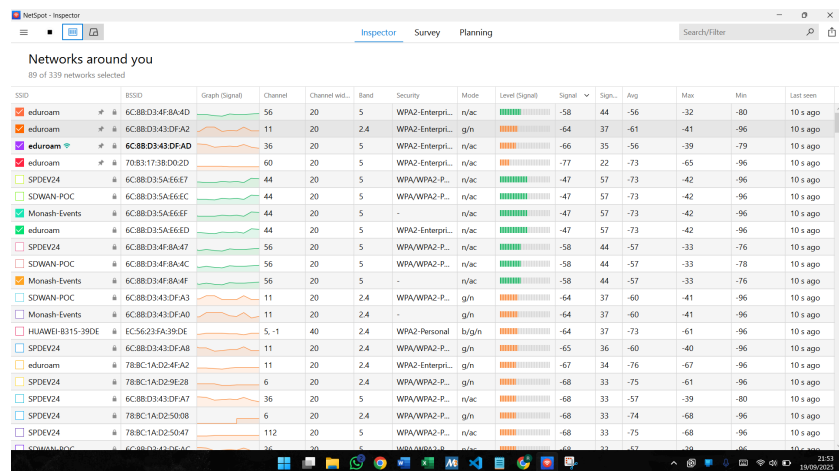
3.2.4. L4



3.2.5. L5



3.2.6. L6



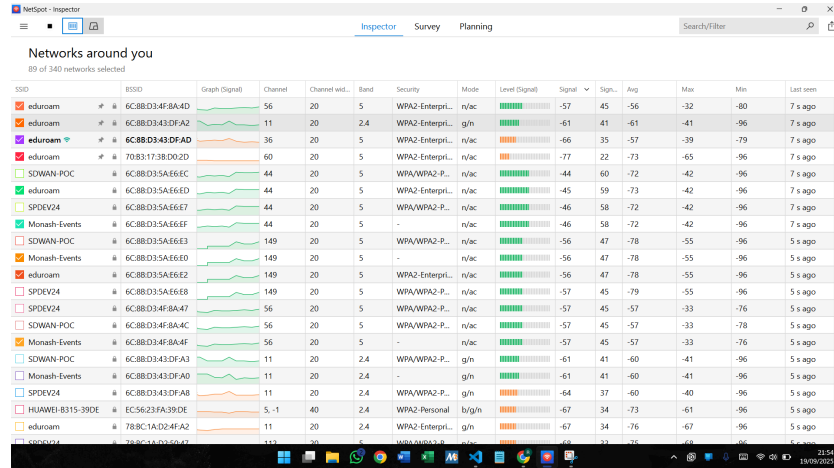
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Networks

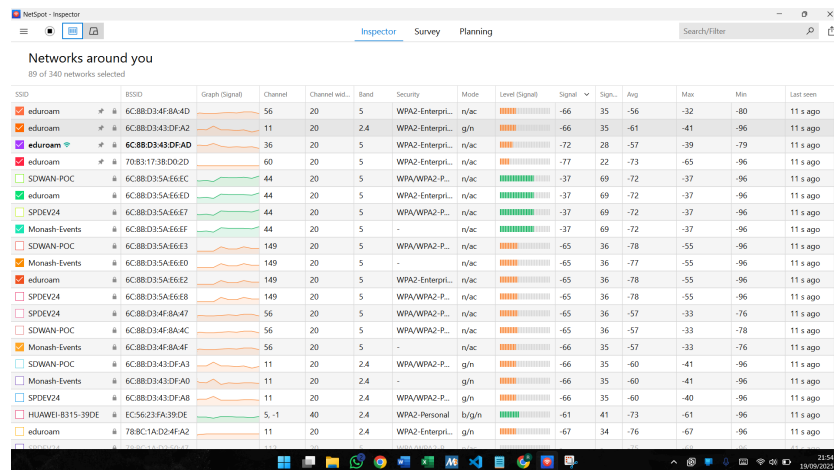
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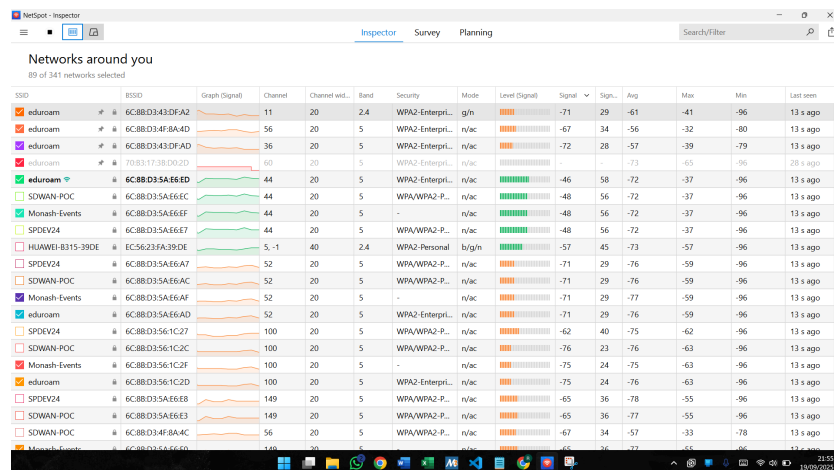
3.2.7. L7



3.2.8. L8



3.2.9. L9



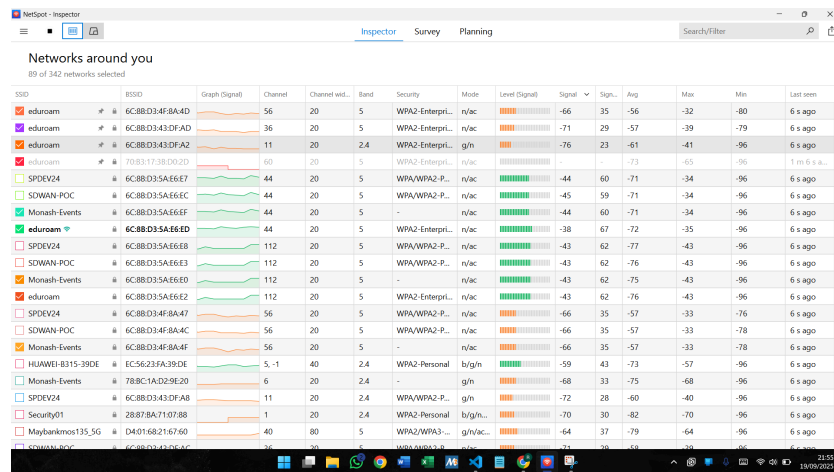
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Networks

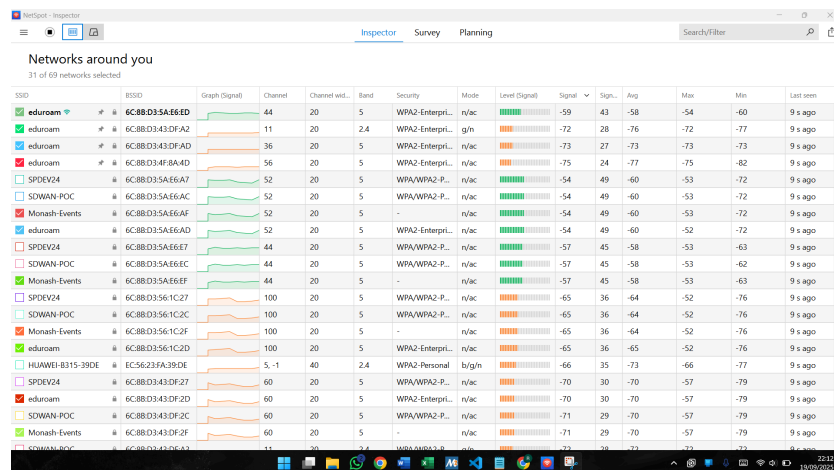
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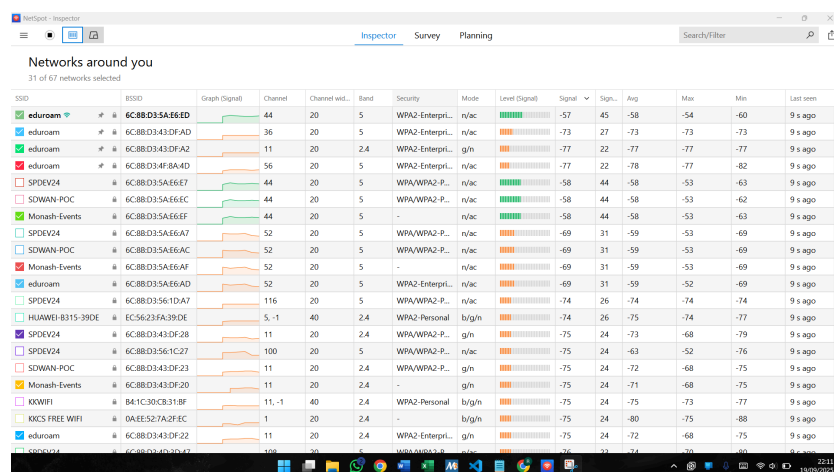
3.2.10. L10



3.2.11. L11



3.2.12. L12



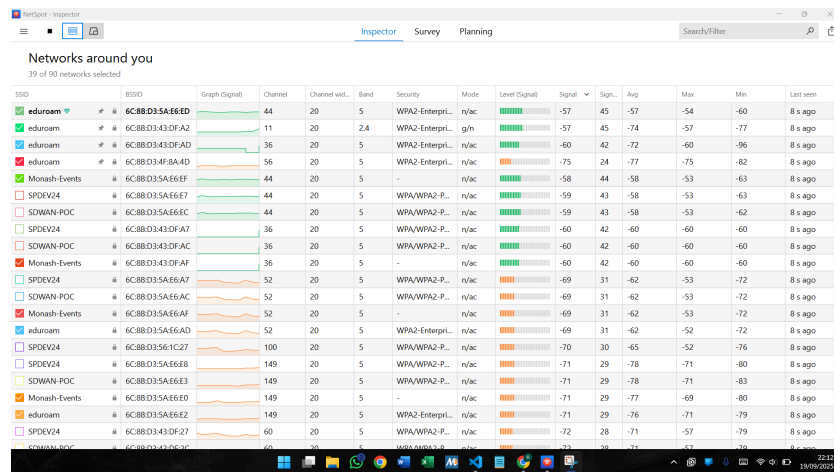
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Networks

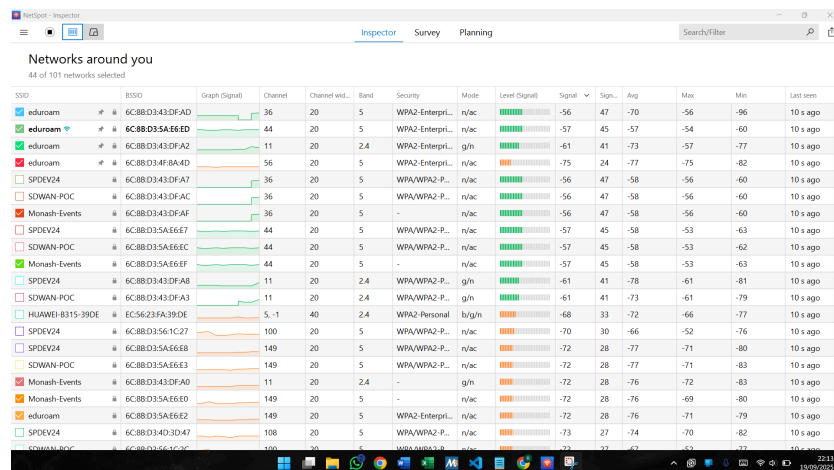
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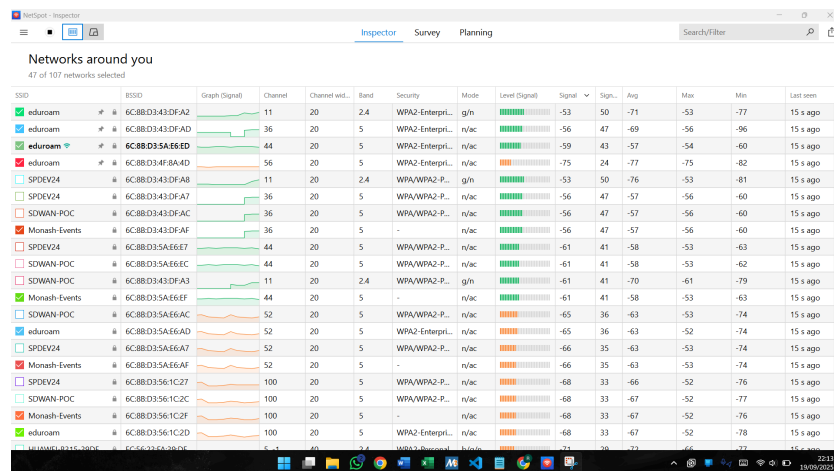
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3.2.14. L14



3.2.15. L15

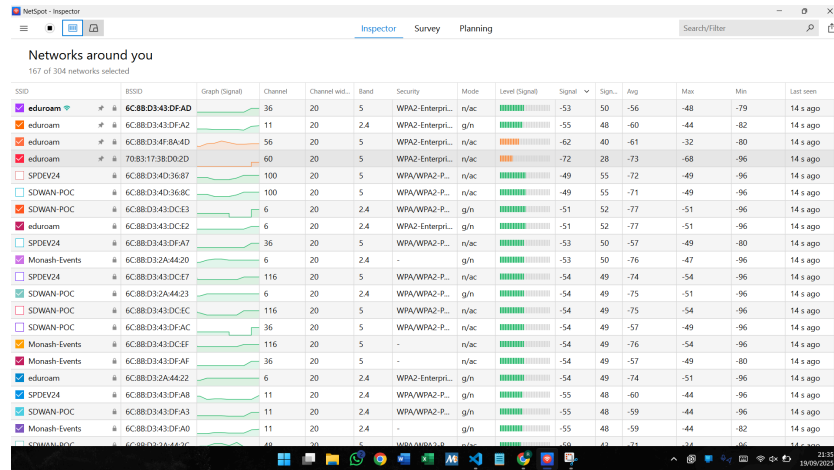


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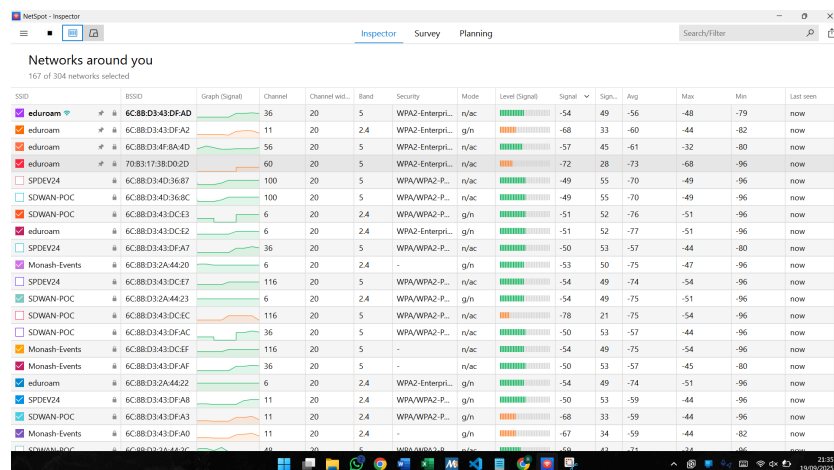
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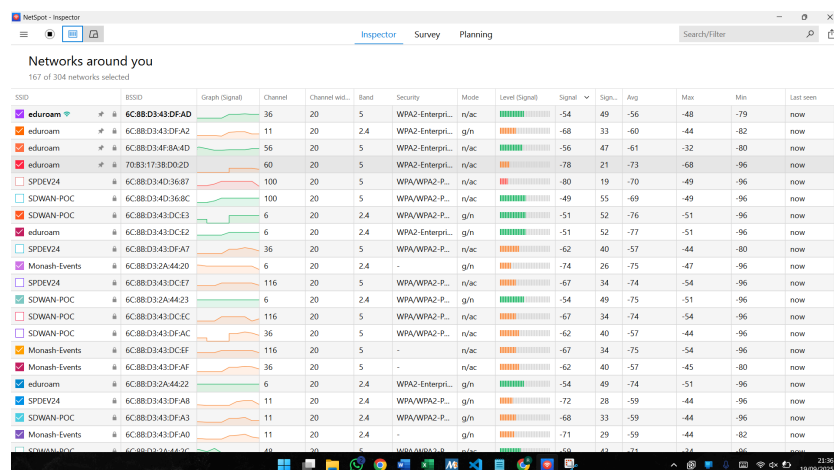
3.2.16. L16



3.2.17. L17



3.2.18. L18



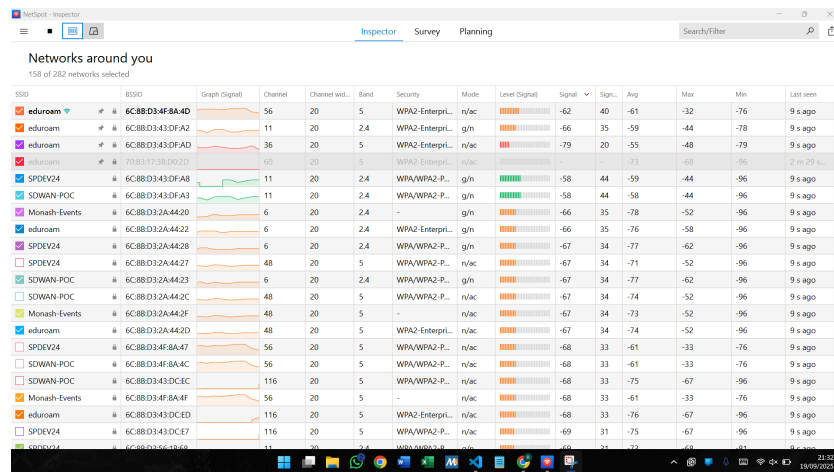
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Networks

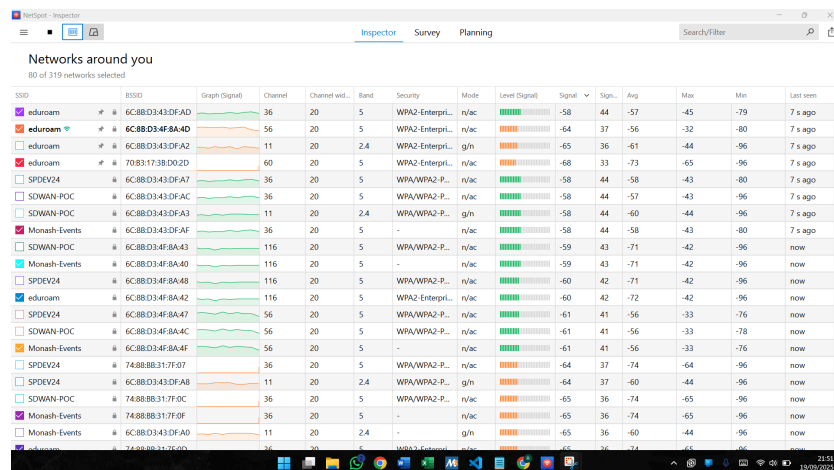
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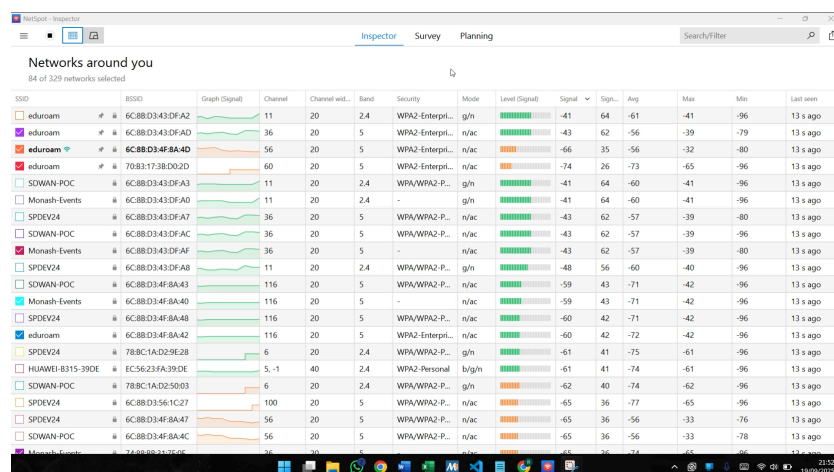
3.2.19. L19



3.2.20. L20



3.2.21. L21



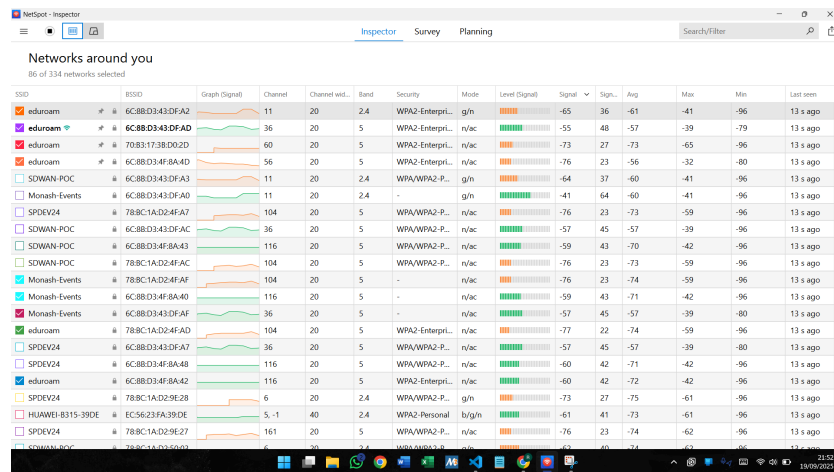
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Networks

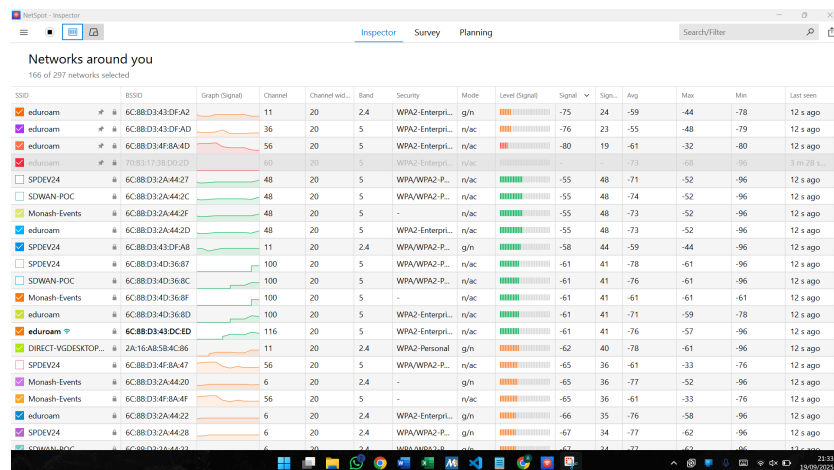
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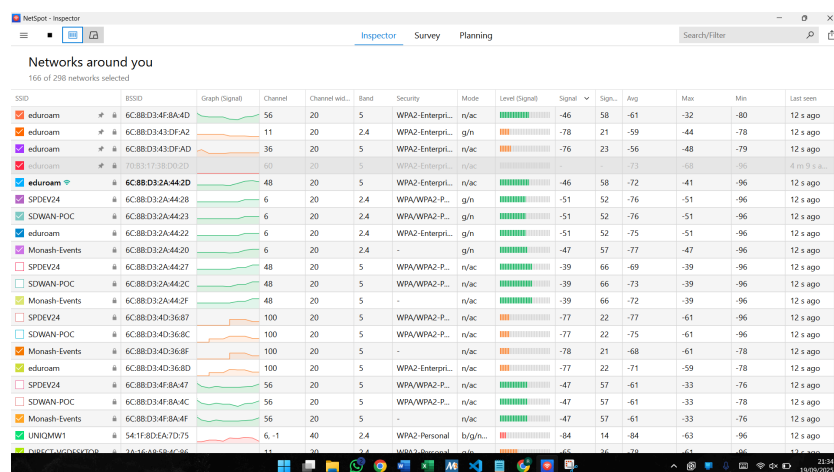
3.2.22. L22



3.2.23. L23



3.2.24. L24



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